



Gaze-based Situational Awareness Estimation System

**FINAL RESEARCH REPORT SUBMITTED
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Abstract:

This project developed a gaze-based situational awareness estimation system for decision-making whilst driving. Research gaps were addressed, which include static knowledge graphs and multimodal integration. This developed system constructs a dynamic knowledge graph from gaze data and scene images to infer a driver's mental map, which then achieves Endsley's SA level 1 (Perception) and partially level 2 (Comprehension). The research question was: how can gaze data and driving-scene context be used to construct a knowledge graph that infers a driver's situational awareness?

Using the Look Both Ways dataset, the system integrates YOLOv8, SAM2, Qwen2.5-VL, and Gemini2.5-Flash. Driver fixations were calculated utilising a dispersion-threshold algorithm. Evaluated on seven different scenarios (70 frames total), the system achieved: YOLOv8 recall of 0.80, precision of 0.97, and F1-score of 0.88; high-quality knowledge graphs from Qwen2.5-VL; and actionable, relevant advice from Gemini2.5-Flash. The results verify VLM integration into a real-time gaze-informed dynamic knowledge graph for SAE.

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1. Introduction

The primary objective of this research project is the development of a gaze-based Situation Awareness (SA) Estimation (SAE) system for decision-making contexts, such as driving. This system will be capable of achieving Endsley's SA Framework Level 1 and may partially achieve Level 2 [1]. SA Level 1 is the Perception of the Elements in the Environment, which includes the location of themselves, their current gaze, and other obstacles/objects within the environment. Level 2 is the Comprehension of the Current Situation, which includes the objects' statuses and relationships between themselves and the user or other objects.

This system utilises eye-tracking data, such as images annotated with fixation points, to evaluate the operator's real-time SA and assess their preparedness for a fallback scenario. This evaluation of the operator is performed by assessing gaze metrics, such as fixation duration and gaze shift frequency. The overarching goal of the system is to be integrated into an In-Vehicle Monitoring System (IVMS) that determines the operator's readiness, such as a coaching or advice system. The research question being asked is the following: How can gaze data and driving-scene context be used to construct a knowledge graph that infers a driver's situational awareness?

This final report presents the entire development and analysis of the gaze-based SAE system. This report will include a thorough literature review, in which research gaps are identified. It will then discuss the methodology, which includes the dataset used, models used, the final approach, and examine the possible risks and ethical concerns. After the methodology has been explained, the report will then explore the results from the system and how it was analysed, and finally end with a conclusion, which includes any recommendations, future works, and key findings.

2. Literature Review

2.1. Introduction

Situational Awareness (SA) Estimation (SAE) has developed into a crucial area of research, specifically using the integration of gaze-based approaches with Artificial Intelligence (AI) and Machine Learning (ML). Accurate SAE is needed for ensuring safe handovers of control in automated vehicles, as a lack of driver awareness or preparedness can elevate the risk of incidents. Therefore, improvements in machine learning and gaze tracking have the capability to improve safety on the road by allowing for real-time evaluation of a driver's attention and readiness. This literature review aims to detect key themes from multiple sources, such as conference papers and journal articles. It also examines similarities and differences between research methods, assesses the strengths and weaknesses of existing research, whilst also recognising gaps in the current research.

2.2. Themes & Gaps

2.2.1. Gaze patterns as SA Measurement

The primary theme discovered in this literature review is the use of gaze patterns and gaze dispersion as measures of SA in automated driving. Some researchers use metrics such as Stationary Gaze Entropy (SGE) to connect eye movement behaviour with SA level 1 during control transition between machine and human. A study discovered that greater SA scores, measured using Situation Awareness Global Assessment Technique (SAGAT), were linked with a broader gaze distribution across Areas of Interest (AOI), with SGE supporting as an estimator of hazard awareness [2]. Another study demonstrated that driving performance was also connected with SA scores [3]. There is an understanding among researchers that non-disruptive capabilities of gaze metrics, with one paper verifying this using a Region-of-Interest analysis [4]. However, another study validates the use of semantic segmentation for modelling peripheral vision, in contrast to static models [5].

Another journal paper [6] reviewed 41 other studies based on eye-tracking metrics for SA. Fixation duration, saccades and gaze direction arose as the important measures, correlating with Endsley's three SA levels. These discoveries confirm gaze patterns as SA indicators. The strengths of this theme are use of validation methodologies, whilst the limitations involved are dependent on controlled experimental environments.

2.2.2. Behavioural metrics for SA

Another key theme focuses on the use of objective behavioural metrics for real-time estimation of SA. One study identified variable saccadic distance, indicating lower SA during manipulation tasks [7]. A different study used a driver response time to decide the SA difference [8]. Similarly, there is an agreement regarding the practicality of objective metrics, with one study showing automated attention detection [9]. However, one study demonstrated high-accuracy object detection [10], which differs with another study that reported inconsistencies [7]. The strengths of this theme comprise non-intrusive validation methods, whilst the limitations involve challenges with data labelling and restrictions in peripheral vision modelling.

2.2.3. LLM to improve SA

The next theme focused on the application of Large Language Models (LLMs) to improve SA, especially in connection with collision risk assessment and handover processes. The Collision Risk Prediction and Takeover Requirements LLM (CPTR-LLM) proposed in one study attained an accuracy of 0.88 in risk estimation, reducing the number of unnecessary handovers [11]. There is also agreement regarding the predictive and informative abilities of LLMs, though one study defines gaps in multimodal fusion [12]. The strengths of this theme include improved understandability, whilst the limitations involve extensive data requirements and a limited scope range.

2.2.4. Multimodal Incorporation for SA

The next theme examines the incorporation of multimodal data with gaze tracking for SAE. The deep learning model presented in one study incorporates flight controls and gaze data, which could achieve an SAE accuracy of 92.04% [13]. Another study linked the handover process to the gaze behaviour [14]. Researchers generally agreed that multimodal fusion improves accuracy; however, one study [13] prioritises eye movement ranking, while another [15] documents that visual alert indicators decreased performance outcomes. The strengths for this theme include the use of AI and ML structures and methodologies, whilst the limitations include computational complexity and the risk of model overfitting due to dependence on simulated data.

2.2.5. Knowledge Graph Incorporation for SA

The final theme discusses the integration of knowledge graphs into SA. One paper [16] created a hybrid Driving Scenario Knowledge Graph (DSKG) using a multi-level ontology and entities and relationships extracted from a dataset. with DistMult exceeding in link predictions. Whilst another paper [17] developed CoSI, a multimodal knowledge graph that integrated driver gaze data with semantic axioms, and achieving 95.3% accuracy in situation classification for SA level 2.

2.3. Literature Review Conclusion

Overall, the literature demonstrates that gaze-based metrics, multimodal fusion, and knowledge graph approaches effectively support SA. However, most knowledge graph frameworks are static and lack dynamic real-time integration. Significant gaps in dynamic, gaze-based knowledge graphs that achieve higher SA levels. Addressing these gaps through the use of SAE frameworks includes the integration of a dynamic knowledge graph combining gaze fixations with object detection, scene context, and AI model-generated relationships.

3. Methodology

As stated in the introduction, the objective of this project was the development of a gaze-based SAE system for driver decision-making. This was concluded after identifying research gaps, such as other studies achieving Endsley's Framework SA Level 1 (Perception) but hardly focusing on Level 2 (understanding) or Level 3 (anticipating). Other important gaps were the use of static knowledge graphs instead of dynamic knowledge graphs and the integration of Multimodal models. These gaps enable a purpose for this project, which was to determine how gaze data and driving-scene context can be used to build knowledge graphs for assessing a driver's SA. From this, a system was proposed to satisfy this research question, of which a diagram can be seen below in Figure 3.1. This diagram includes the inputs, outputs, and a possible application of this system.

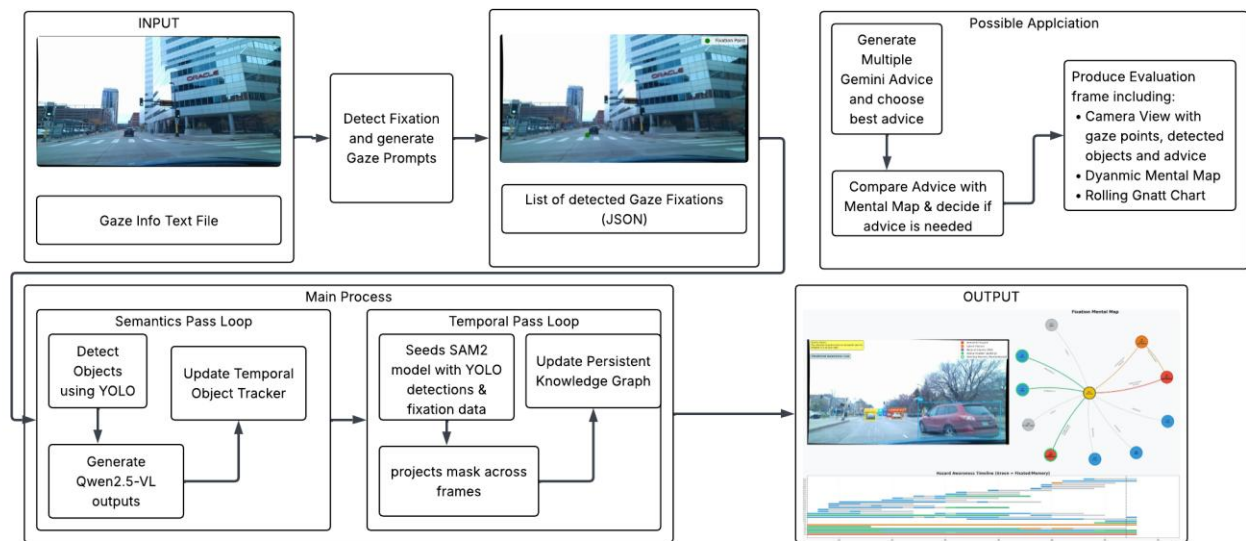


Figure 3.1: Proposed System Design

The inputs of this system would be an image of the scene ahead of the car, and the gaze data, which would include the 2D coordinates of the driver's gaze that correspond to the image. The expected outputs of this system are the image, with detected objects and classifications, and the corresponding knowledge graph that shows a driver's mental map, which should include the objects and their relationships and statuses. This system is intended to achieve Endsley's framework SA level 1 and SA level 2, whilst providing a dynamic knowledge graph. A possible application of this is also seen in the top right diagram, which is a driving coach.

3.1. Dataset

Several datasets were suggested by the Supervisor, including Look Both Way (LBW) [18], EyeCar [19], DADA-2000 [20], and Dreyeve [21]. After analysis and discussion, the LBW dataset was selected. It comprises data from 28 subjects, with each subject having 2 or 3 different scenarios. The dataset consists of 300GB worth of data, with around 7 hours of driving data in total. These scenarios include different environments (residential, rural, and urban), weather conditions (cloudy, snowy, sunny, and, rainy), and time of day (day or night) with each subject, and there are many driving scenarios. There are four different folders:

- scene_ims: images taken from a forward-facing camera showing what is in front of the vehicle. (e.g. 00000027_ims.png)
- scene_depth: allow for the depth of the image to be analysed along with the image. (e.g. 00000027_depth.npy)
- face_ims: images taken from a backward-facing camera showing the subject's face. (e.g. 00000027_face.png)
- gaze_info: Text files containing 2D/3D gaze data and eye locations relative to the scene (00000027_info.txt)

The only folders that are used from this dataset are the 'scene_ims' folder containing the forward images and the 'gaze_info' folder containing the gaze data. All files are time-synchronised via matching time stamps. Unfortunately, this dataset was reduced as the original capture frequency was 15Hz but was downsized to 5Hz for processing [22].

3.2. Models

3.2.1. Segmentation: SAM2

No specific models were recommended for segmentation. Through research, Meta's Segment Anything Model (SAM) was selected. SAM is an image segmentation framework and has been widely applied across diverse tasks, including scene and object level segmentation, and in various environments such as virtual reality, the medical field, and, in particular, automated vehicles.

SAM solely accepts images as its input, thus limiting its application. SAM2 is an enhanced version that supports both image and video inputs. SAM was specifically optimised and designed to efficiently power its large-scale data engine. The SAM disconnects its framework into a one-time image encoder and a mask decoder, which can both execute in a few milliseconds per input [23]. The SAM2 operates identically to SAM for images; however, it changes its approach when processing video inputs. The SAM2 includes a memory attention process between the encoder and decoder, along with a memory bank at the end of the pipeline. A visual representation of how each model works can be seen in Appendix I, and justification for why SAM2 was used for this project is seen in Appendix B.

3.2.2. Object Detection: YOLO

No specific models were recommended for object detection. The 'You Only Look Once' (YOLO) model was selected, primarily due to the team members having prior experience with it. The YOLO model is a widely recognised framework, primarily used for real-time object detection within an image or video, and known for its high accuracy and computational efficiency. [24]. The YOLO models utilise their own YOLO algorithm built upon a Convolutional Neural Network (CNN). The process begins by dividing the image into a grid of cells. Each cell then generates several bounding boxes. Each bounding box is bound by four coordinates corresponding to the corners of the box.

Each bounding box is also accompanied by a confidence score representing the likelihood of the box containing an object. Each cell also predicts the likelihood of a detected object belonging to a set of predefined classes. After processing each cell, the bound box coordinates, confidence score, and classification probability predictions are compiled together to produce a final output. The final process includes Non-Maximum Suppression, which is used to eliminate redundant overlapping bounding boxes that detect the same object [25].

The selected YOLO model is a pre-trained model trained on driving/traffic data and can be presented in a Medium Article [26]. This version will enhance object detection performance, as it has been trained specifically on traffic and driving data. In contrast, general YOLO models are trained on a wide range of datasets meaning they tend to detect a wide variety of irrelevant objects. The justification for selecting the YOLO model is provided in Appendix C.

3.2.3. Object Relationship VLM: QWEN

The supervisor suggested several examples, such as Meta Llama 3.2 and Google Gemini. Later in the project, it was recommended to explore Alibaba Cloud's Qwen model, which proved to be a suitable choice for project integration. Qwen is a family of Large Language Models (LLMs) developed by Alibaba Cloud. These models are proficient in understanding and processing both textual and multimodal data, including images and videos. Some are designed for text analysis and creation, while others are designed for image processing [27]. The Qwen2.5-VL-7B variant was selected as it was the most up-to-date version of the Qwen2-VL Vision Language Model (VLM). The Qwen2.5-VL-7B is specifically designed to accept both text and images as inputs and to create outputs in both textual and visual formats.

The Qwen2.5-VL model operates through three core components: an LLM, a vision encoder, and a Multi-layer Perception (MLP)-based Vision-Language Merger. The LLM component within the VLM is the Qwen2.5 LLM, which has been modified to better accommodate the required inputs. The vision encoder utilises a modified Vision Transformer framework that incorporates 2D Rotary Position Embedding (2D-RoPE) and window attention. This preserves native input resolutions while accelerating encoder computation speeds. The MLP-based Vision-Language Merger is utilised to compact the image features and related data before passing them to the LLM [28]. The justification for selecting the Qwen2.5-VL model can be seen in Appendix D.

3.2.4. Driver Advice and SA Score VLM: Gemini

The supervisor suggested using Google Gemini in particular for this task. Gemini is Google's main family of AI. It is multimodal, meaning it is designed to comprehend and create text, audio, images, and video [29]. The model of Gemini that will be used is the Gemini-2.5-Flash, a fast and lightweight version that allows for a balance between speed and accuracy. The justification for selecting the Gemini2.5-Flash model can be seen in Appendix E. Considering that Google uses these Gemini models in applications, such as Google Search and Google Maps, this would significantly help when it comes to driving advice, as the model has high-volume inference. With its current applications and with the scene images and gaze data included with an instruction prompt, this shows that Gemini would be a very suitable AI model to use in the case of a driving coach.

3.3. Final Approach

Many conceptual stages occurred before this final approach was envisioned, in which these conceptual stages can be seen in Appendix F, G and H. This final approach draws heavy inspiration from conceptual stage 3 and that stage took borrowed findings from a paper called ‘A Framework for Leveraging LLMs for Scene Analysis and Cognitive Processing’ [30]. This approach follows a 5-step process, concluding with a possible application stage. This pipeline process can be seen below in Figure 3.2.

- Step 1). Set up and Data Loading
- Step 2). Gaze data processing, fixation detection, mapping fixation to images and generate gaze prompts for segmentation
- Step 3). Model initialisation (pre-train YOLOv8, SAM2, Qwen2.5-VL and Gemini)
- Step 4). Semantic Pass Loop
- Step 5). Temporal Pass Loop

Figure 3.2: Final Method Process

The first step includes initial setup and data loading. This involves installing the necessary Python libraries and toolkits (OpenCV), mounting the Google Drive containing the filtered subject’s driving dataset, and defining paths to the relevant folders.

The second step focuses on processing gaze data and detecting fixations. Gaze data is extracted from text files, followed by applying the Dispersion-Based Algorithm (I-DT) for fixation detection, based on the pseudocode from ‘Identifying Fixation and Saccades in Eye-Tracking Protocols’ [31]. This algorithm scored the highest overall in the referenced study and uses dispersion and duration threshold as inputs. After reviewing more papers and studies on fixation duration [32] [33] [34], a threshold of 100ms was selected as a fair value. The step outputs a list of detected fixations, including start and end frames, position coordinates, and fixation duration. This step also maps each fixation point onto its corresponding image, marked via a green dot. An example of this is provided in Appendix J. This step also generates gaze prompts for the later segmentation section by transforming the mapped fixation data into JSON format. This serves as the necessary input for SAM2. The output is a list including all mapped fixation results.

The third step loads the required models that will be used for later steps. This includes initialising the pre-trained YOLOv8 for object detection, the SAM2 checkpoint for segmentation, the Qwen2.5-VL, and the Gemini API. Qwen2.5-VL is employed for tasks such as graph generation and scene description, while Gemini is utilised for providing advice and performing SA scoring. Additionally, the ‘TemporalObjectTracker’ is initialised to sustain consistent object IDs across frames, and the ‘PersistentKnowledgeGraph’ is set up to maintain a clear understanding of objects, their status, and correlation over time, even when objects temporarily leave the field of view.

The fourth step is the Semantic Pass Loop (SPL), which forms an important section of the pipeline, allowing for individual frame analysis. This is crucial as it improves the system's understanding by allowing for semantic reasoning. In each frame, it first performs object detection using the YOLO model to identify all relevant objects in the frame. These detections, along with the original image, are then given to the Qwen2.5-VL VLM. Qwen-2.5 then performs a detailed semantic analysis by sorting the detected objects into three categories with varying priority. Imminent are objects posing an immediate threat. Latent objects are those that have the possibility of becoming hazardous in the future. Points of Interest (POI) are objects that are not currently critical but still require monitoring. The prompt used for semantic analysis can be seen in Figure 3.3, showing the three categories the objects could be.

```
prompt_triage = """Classify important objects into ONE of three categories:
1. imminent: Immediate hazards.
2. latent: Potential future hazards.
3. poi: Points of interest.
Respond ONLY with a comma-separated list like this: T0: imminent, T2: latent. If none
apply, say NONE."""
```

Figure 3.3: Semantic Analysis Prompt for Qwen2.5-VL

Additionally, Qwen-2.5-VL also creates contextual descriptions of spatial and semantic relationships, involving primary and secondary relationships. A primary relationship is a correlation between the ego-driver and the objects surrounding them, such as ‘the truck is driving ahead of the ego-driver’. A secondary relationship is a correlation between one object and another possible object, such as ‘a traffic sign is to the right of the truck’. The prompts used to identify these relationships can be seen below in Figures 3.4 and 3.5. This semantic relationship information is used to update the ‘TemporalObjectTracker’, assuring object identification is consistent across all frames analysed.

```
prompt_primary = f"""Look at THESE SPECIFIC tracked objects currently in the scene: {tids_str}.
Describe what each object is doing and its location relative to the camera car (EGO).
(e.g., driving ahead, oncoming traffic on left, parked on right).
Keep it under 5 words per object.

OUTPUT FORMAT:
REL: EGO->T0: parked on right
REL: EGO->T5: oncoming on left
"""
```

Figure 3.4: Primary Relationship Prompt for Qwen2.5-VL

```
prompt_sec = f"""Look at THESE SPECIFIC tracked objects: {tids_str}.
Identify if any of these objects are near EACH OTHER (not the EGO car).
If they are near another object, then describe their relation to each other.
(e.g., T2 following behind T3).
If none are near others, say NONE.

OUTPUT FORMAT:
REL: T2->T3: following behind
REL: T5->T8: parked next to
"""
```

Figure 3.5: Secondary Relationship Prompt for Qwen2.5-VL

The fifth step is the Temporal Pass Loop (TPL), which focuses on ensuring temporal consistency and updating the system's overall scene understanding. This is essential for tracking object paths, predicting possible relations, and supporting SAE across long sequences. In this step, the SAM2 model is seeded with the object detections and driver's gaze fixation data for the current frames. It then creates these object masks across successive frames in an out-of-memory (OOM) safe manner, allowing for temporal consistency even when objects momentarily leave or are obscured from the driver's view. At the same time, the TPL updates the ‘PersistentKnowledgeGraph’ with the new information. This update to the ‘PersistentKnowledgeGraph’ includes processed object details from the YOLO model, semantic classification, and relationships from the Qwen-2.5-VL VLM and the driver's current fixation. By constantly including this input data, the ‘PersistentKnowledgeGraph’ ensures an understandable, evolving representation of the dynamic driving environment over time.

3.4. Application

The possible application of the system proposed is the driving coach, which would give advice to drivers based on their SA. This driving coach would employ the Gemini-2.5-VL-Flash model to create quickly generated, brief advice based on the current scene understanding and the updated 'PersistentKnowledgeGraph'. This advice is chosen from a selected number of pieces of advice based on the same image. Then another Gemini prompt is used to choose the best advice to be used. The prompts used for advice generation and the advice selection can be seen in Figures 3.6 and 3.7 below.

```
current_instructor_prompt = textwrap.dedent(f"""
    You are an experienced driving instructor sitting next to a novice driver.
    Examine the current driving scene carefully. Provide short, concise advice
    to the driver, highlighting any potential risks, conflicts, or important
    observations. Focus on what the driver *should pay attention to* or *act upon* (E.G.
    Missed Hazard: Truck on Left).
    Be direct and specific. Respond with a single sentence, less than 25 words.
    Do NOT include newlines or excessive whitespace in the advice.
    Your response MUST be in JSON format, like this example:

    ```json
 {{
 "frame_id": "{idx}",
 "gemini_advice": "Missed Hazard: Car on the Right."
 }}
    ```
    """)
```

Figure 3.6: Advice Generation Prompt for Gemini2.5-Flash

```
selection_prompt_text = textwrap.dedent(f"""
    You are a senior driving instructor. Given the current driving scene and multiple candidate
    pieces of advice, select the *single best*
    and most actionable advice for the driver.
    Consider clarity, directness, and relevance to potential hazards. The selected advice
    should be a single sentence, less than 25 words.
    Do NOT include newlines or excessive whitespace in the advice.
    Your response MUST be in JSON format, like this example, containing only the selected
    advice:

    ```json
 {{
 "frame_id": "{idx}",
 "gemini_advice": "Missed Hazard: Car on the Right."
 }}
    ```

    Candidate Advices:
    {chr(10).join([f"Advice {i+1}: \"{advice}\"" for i, advice in
    enumerate(filtered_candidate_advices)])}
    """)
```

Figure 3.7: Advice Selection Prompt for Gemini2.5-Flash

It then performs the SA scoring by assigning a score to the driver's awareness level as low, medium, or high. Low is shown when the driver is considered to be not aware of hazards and POIs, and driver gaze is misdirected from these hazards. Medium is displayed when the driver is aware of some hazards and POIs, and the driver's gaze is distracted intermittently. A driver is aware of most hazards, and the driver has acknowledged these hazards mentally and visually via the driver's gaze.

It then performs the SA evaluation by comparing Gemini advice against the current knowledge graph to identify which objects are mentioned in both. Then, a decision is made whether the advice generated should be displayed or not. The condition for displaying the advice is that if the advice mentions a critical object that is not contained in the mental map, or the advice contains words indicating a hazard that needs attention. Finally, all the outputs from the one loop iteration are compiled into a single image. The contents of this image can be seen in Figure 3.8 below. Once complete, these evaluation frames are compiled again into a video format for overall scene evaluation.

- Camera View: includes the frame with detected objects, their labels and statuses, SA score and when triggered the Gemini advice.
- Mental Map: knowledge graph of primary and secondary relationships and their statuses.
- Gantt Chart: shows driver attention and object lifecycle.

Figure 3.8: Content after System is applied the application.

3.5. Risks & Ethics

3.5.1. Risk

Several situations have occurred where key team members or the supervisor were unable to attend scheduled meetings due to other commitments or the inability to meet in person. This risk has been effectively managed by immediately organising backup meetings when all relevant personnel are available or entrusting other personnel to be brought up to speed by others. This risk was recognised and foreseen to happen, which is why communication between personnel was key when it came to mitigating this risk. The dataset used is filtered but complete, as it still contains a representative version of the collected data. No data privacy breaches have occurred as the LBW dataset is publicly available and already anonymised the participants through subject numbering. However, a remaining consideration was that certain locations may be recognisable, potentially allowing others to conclude where the participants are located. Another risk that occurred was that updating certain programming libraries may alter or destroy the code's normal functionality, creating major problems for a safety system. This risk could be mitigated by using version pinning, making sure the libraries are utilised using a specified version.

3.5.2. Ethics

This table defines the ethical considerations applied throughout the project when using the LBW dataset. As shown in Table 3.1 below, it maps specific clauses from the Engineering Australia Code of Ethics and Guidelines on Professional Conduct to the corresponding ethical measures implemented in this project.

Table 3.1: Consideration of Engineering Australia's Ethic Guidelines

Clause	Focused Area	Consideration
Clause 1: Demonstrate Integrity	Public Interest and Safety	When using the LBW dataset, which contains sensitive operator and scene data, priority is given to public safety by ensuring all evaluations contribute to enhanced road safety. Subject anonymity is maintained through the use of numerical identifiers, and facial images will be discarded if not necessary. This approach complies with 1.1, 1.2, and 1.3 Clauses within the guidelines. [35].

Clause 2: Practise Competently	Competence and Professional Development	Competence was upheld by selecting well-proven models, including YOLO, SAM2, Qwen2.5-VL, and Gemini. These models were validated through testing on the LBW dataset to ensure accurate segmentation and object detection. This project adheres to the 2.1, 2.2, and 2.3 Clauses within the guidelines [35].
Clause 3: Exercise Leadership	Team Collaboration & Ethical Guidance	Team collaboration occurred regularly through in-person and online meetings with the supervisor and their associates. These meetings included discussions, evaluations, and finalisation of the dataset, models, and overall project approach. This project adheres to the 3.1, 3.2, and 3.3 Clauses within the guidelines [35].
Clause 4: Promote Sustainability	Sustainable Development and Societal Impact	Sustainability was addressed by employing efficient computational models to minimise resource usage on Google Colab. Socially, the project enhances driving safety using the LBW dataset to support future well-being. Economically, it relies on open-source models (YOLO, SAM2, Qwen, and Gemini). This project complies with the 4.1, 4.2, and 4.3 Clauses within the guidelines [35].

4. Results, Analysis & Discussion

4.1. Results

The results given by this pipeline come in multiple formats, such as a summary video, a 'JSON' file containing all the advice for each frame, and a folder containing the frames that make up the summary video. For this project, 7 different scenarios, 10 frames from each (70 altogether) were tested, with each having different environments, weather conditions, and time of day. These seven scenarios and their details can be seen below in Figure 4.1, which shows that most scenarios took place during the sunlight hours with no crazy weather. There are two outliers in these scenarios, which are Subject 19-1 and Subject 26-2. An example frame of what is produced can be seen in Figure 4.2 below.

- Subject 06-2: Normal, Urban, Sunny
- Subject 11-1: Normal, Urban, Overcast
- Subject 17-3: Normal, Urban, Glare
- Subject 19-1: Snowy, Residential
- Subject 24-2: Normal, Urban, Overcast
- Subject 26-2: Nighttime, Urban
- Subject 28-2: Normal, Residential, Overcast

Figure 4.1: Scenarios Tested

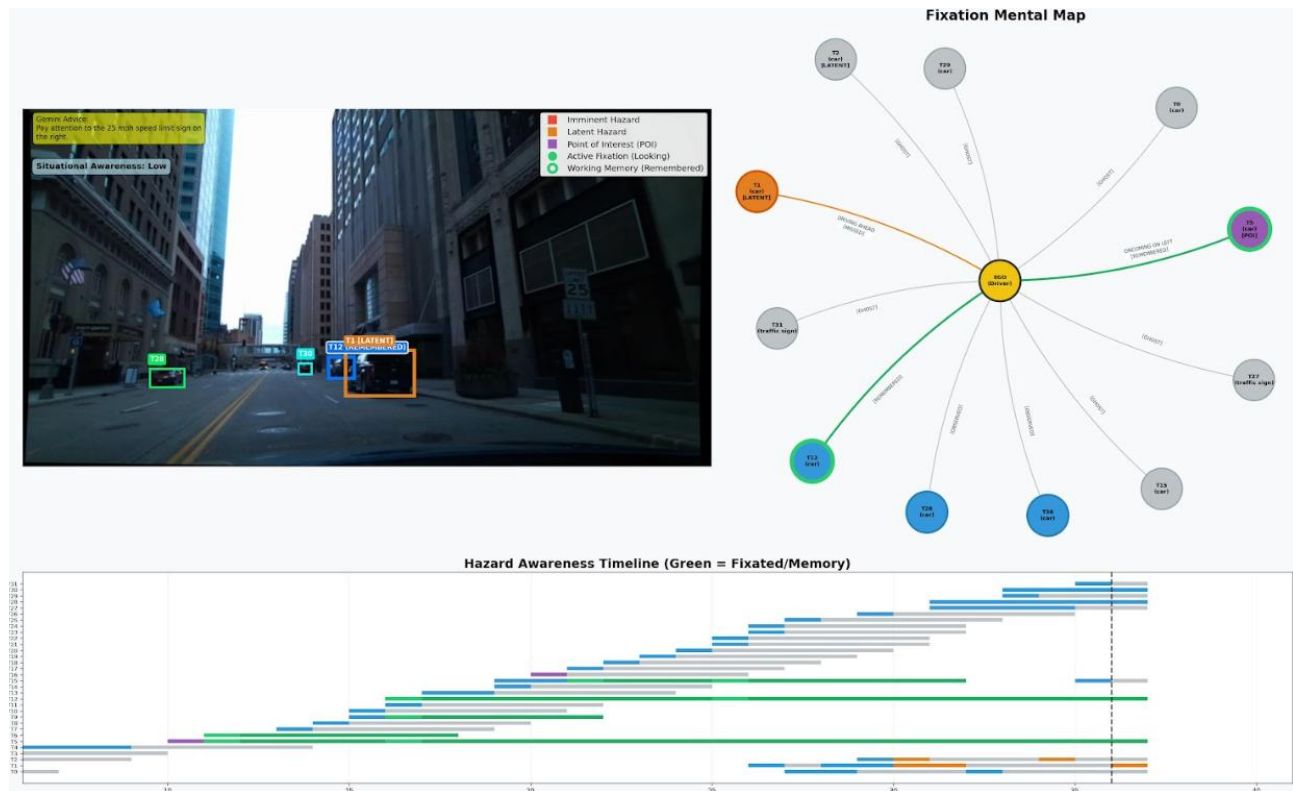


Figure 4.2: Example of Final Output with driving advice

An example frame of what is produced can be seen in Figure 4.2 above. This frame shows the image, the knowledge graph or mental map, and the Gantt chart of events. The image shows the detected objects with bounding boxes that include the mental map identifier along with its current fixation status and a legend to understand what different coloured bounding boxes are. The image also includes the Gemini component, with the advice stating an important object the YOLOv8 has missed, and Gemini considers the driver to have a low SA score. The knowledge graph is shown to the right of the image showing the driver's mental map, each object being classified, and relationships between the driver and objects and objects and other objects. Each edge/relationship in the mental map includes the contextual relationship somewhere along the relationship line. Finally, at the bottom of the page is the Gantt chart, where along the bottom are the frame numbers and along the side are the object IDs. The code for this project can be seen in Appendix J. Evaluation frames for the 70 selected frames can be seen in Appendix L.

4.2. Analysis

The analysis of this project and its models was split into separate sections, each evaluating a model or models that are important parts of the project. Overall, the model performed very well, considering the mid-project problems that occurred. The project was able to roughly estimate a driver's SA and advise the driver of hazards he may be unaware of. The full spreadsheet for analysis can be found in Appendix K, which includes all of which will be discussed below

4.2.1. YOLOv8 Object Detection

For evaluating the object detection, the decision was made to use machine learning classification metrics, such as recall, precision, and F1-score. To do this, manual analysis of frames to determine the ground truths was undertaken, where the number of missed detections, wrong detections, and correct detections was discussed and agreed upon. The number of missed detections was considered the false negatives, the number of wrong detections was considered the false positives, and the number of true positives was considered the correct detections. The evaluation metric for object detection can be seen in Table 4.1 below.

Table 4.1: Object Detection Evaluation Metrics

Scenario	Recall	Precision	F1-Score
Subject 06-2	0.784	0.952	0.86
Subject 11-1	0.727	1	0.842
Subject 17-3	0.813	1	0.897
Subject 19-1	0.809	0.974	0.884
Subject 24-2	0.854	0.976	0.911
Subject 26-2	0.464	1	0.634
Subject 28-2	0.958	0.958	0.958
Overall	0.796	0.971	0.875

As seen above, the pipeline processes were mostly above the average overall score for their metrics, the overall recall being 0.796, the overall precision being 0.971, and the overall F1-score being 0.875. The lowest recall was 0.464 from Subject 26-2, and the highest was 0.958 from Subject 28-2. The lowest precision score was 0.952 from Subject 28-2, and the highest was 1 from Subject 11-1, Subject 17-3, and Subject 26-2. The lowest F1-score was 0.634 from Subject 26-2, and the highest was 0.958 from Subject 28-2. One anomaly for this evaluation is Subject 26-2, which had a perfect precision but a very low recall, thus lowering the F1-score to a low quality. This anomaly was the nighttime scenario, and the scores suggest that the model can perfectly classify the objects that are detected but fails to detect all objects within the image. This is most likely due to hardware limitations or model augmentations such as thresholds. Either way, this is a single scenario that should be improved upon.

4.2.2. Qwen2.5-VL VLM

For evaluating the Qwen2.5-VL VLM, it was decided that a human quality evaluation had to be done, as there was no numerical data to evaluate from the VLM outputs. Instead, the knowledge graph was manually evaluated by two team members, as ground truths would not exist for this. The metrics that were used in the end were Very High, High, Medium, Low, and Very Low. Very High is given if the knowledge graph had no mistakes. High is given if the knowledge graph had one mistake. Medium is given if the knowledge graph had two or three mistakes. Low is given if the knowledge graph has between four and six mistakes. Very Low is given if the knowledge has seven or more mistakes. The average result for each scenario can be seen in Figure 4.3. (A more detailed version can be seen in Appendix K)

- Subject 06-2: Medium

- Subject 11-1: Medium
- Subject 17-3: Medium
- Subject 19-1: Medium
- Subject 24-2: High
- Subject 26-2: High
- Subject 28-2: Medium

Figure 4.3: Scenario average metric for mental map quality

As seen above, the Qwen2.5-VL was very good at delivering quality mental maps of the drivers' SA, with the VLM usually making a few mistakes, sometimes only making one mistake, and rarely making more than four mistakes. Although Subject 26-2 had an average mental map quality of high, this may be due to the later quarter of the dataset, where the driver is stuck at an intersection, meaning that the mental map will only update when an object passes them. An example of this possibility can be seen in Figure 4.4. Another problem that was discovered with the VLM was that it could be overloaded, and the knowledge graph relationships could go haywire, such as an object having a relationship to itself, which can be seen as an example in Appendix M.

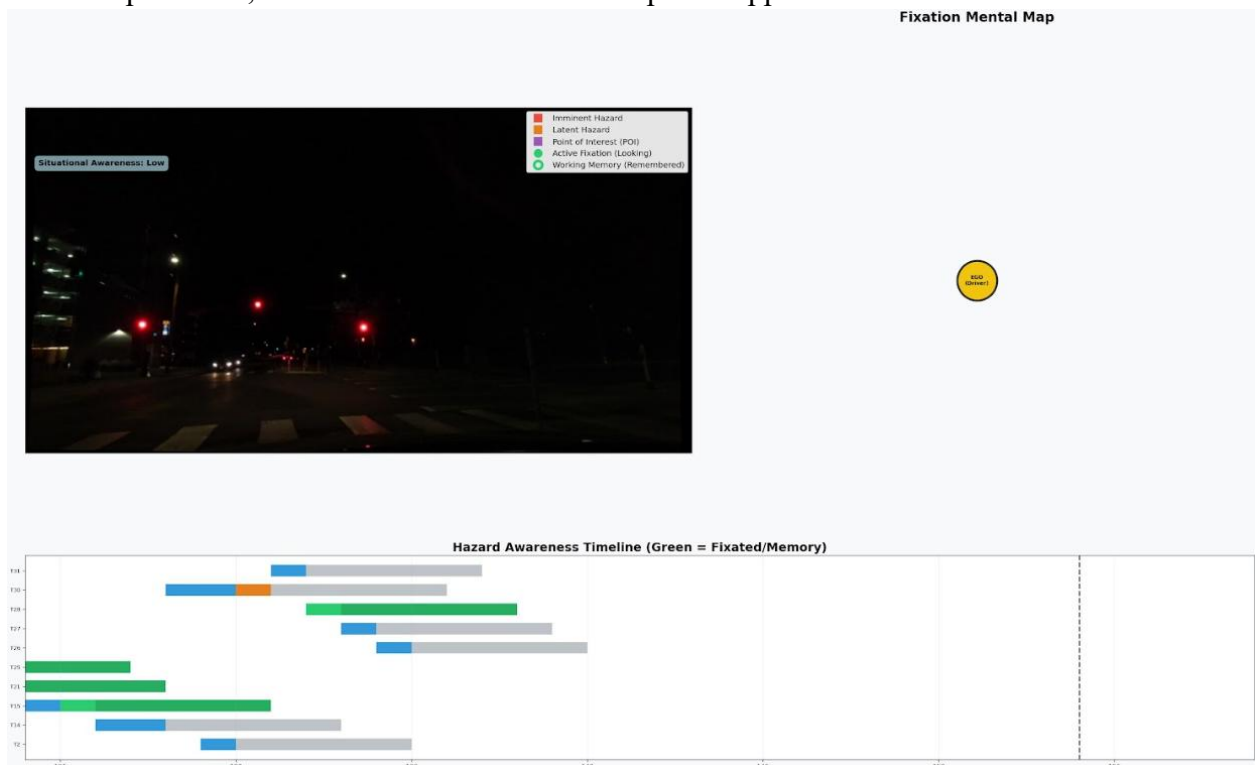


Figure 4.4: Frame Evaluation for Subject 26-2, frame 154

4.2.3. Gemini-2.5-flash

A similar approach was used for the Gemini model evaluation, as there are no ground truths given. The metrics that were used for this evaluation are Very Good, Good, OK, Bad, and Very Bad. Very Good is given if the Gemini advice is relevant and useful. Good advice is given if the Gemini advice has missed 1 piece of advice. OK is given if the Gemini advice has partial information and missed some things. Bad is given if the Gemini advice is misleading or irrelevant. Very Bad is given if the Gemini advice is misleading enough to possibly cause an accident. The average result for each scenario can be seen in Figure 4.5. (A more detailed version can be seen in Appendix K)

- Subject 06-2: Good
- Subject 11-1: Good
- Subject 17-3: OK
- Subject 19-1: Good
- Subject 24-2: Good
- Subject 26-2: Good
- Subject 28-2: Good

Figure 4.5: Scenario average metric for Gemini Advice quality

As seen above, the Gemini advice was excellent at delivering good and reasonable advice to the driver, with the Gemini model usually making one mistake, rarely giving wrong or irrelevant advice, and never giving accident-possible advice. One good example of how the Gemini advice is used can be seen in Subject 26-2, where the Gemini has picked up on something the YOLO and Qwen2.5-VL model have missed, spotting a pedestrian to the right of the screen just entering onto the crossing. This example can be seen below in Figure 4.6.



Figure 4.6: Camera image from Frame Evaluation for Subject 26-2, Frame 97

Although the advice was mostly accurate and relevant to the image the Gemini model was given, there were instances where the Gemini model was completely wrong. An example of this is seen in Figure 4.7, which shows advice saying there is a car to the right when, in fact, there is no car to the right. It is unknown why the model gave this advice, but it may be due to the prompt to see the car behind the fence.

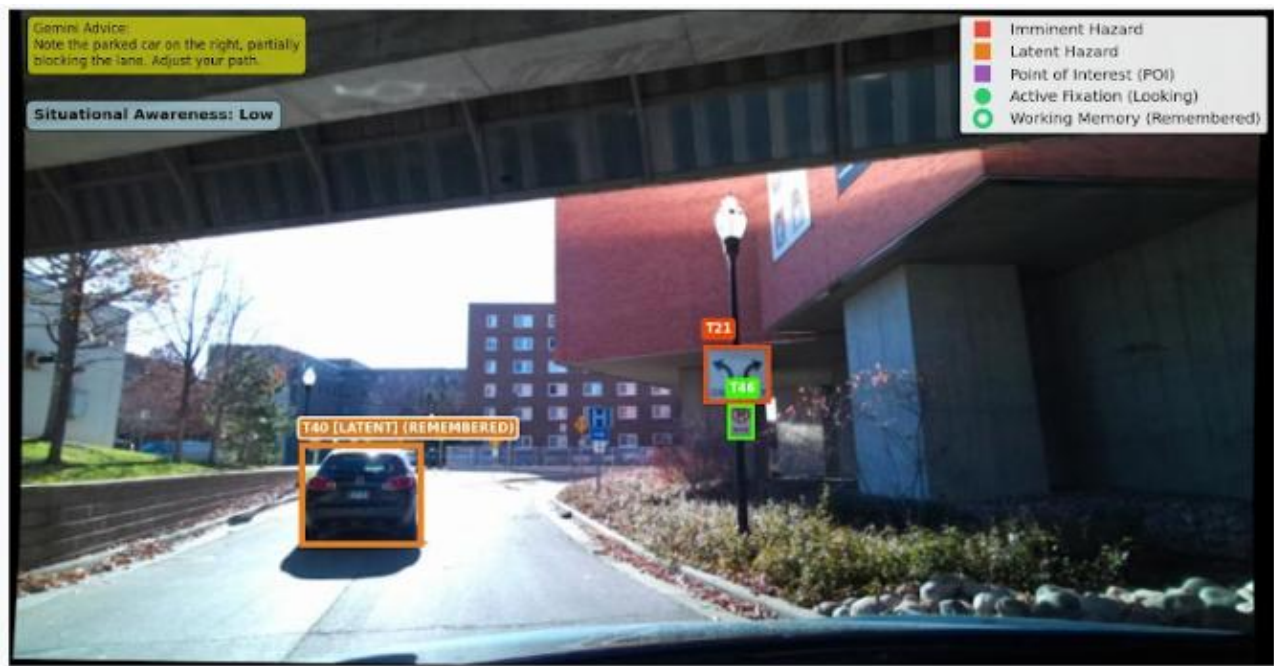


Figure 4.7: Camera image from Frame Evaluation for Subject 17-3, Frame 151

4.3. Risks, Ethics & Limitations

The main risk that kept occurring was the availability of key personnel to be available for meetings, which was easier to combat, as nearly all meetings were online, except for four or five. The one major risk was the unfortunate update to some of the Python Libraries used in previous stages of development, affecting the generation of knowledge graphs and use of SAM2. This recovery from this risk took the third quarter of the project to be able to display correct outputs again.

No ethical codes of the Engineering Australia Code of Ethics and Guidelines on Professional Conduct were broken, as all the considerations from the previous part of this report were upheld. Although the dataset contains the subject's face, the faces aren't used during the pipeline, thus allowing for anonymity to be maintained. This project used some of the most advanced and up-to-date models, and evaluations on these models were undertaken to ensure this professionalism. Many meetings were held where parts of the project were discussed, analysed, and finalised by the supervisor and key personnel. Sustainability was upheld by using open-source models and socially, as this project is to enable safer driving. There was the possibility of using it in a driving simulator with participants, which would need approval from the University Human Research Ethics Committee Charter (UHREC) as participant faces would need to be used for gaze data. Due to the major risk that occurred, it was decided that more resources should be put towards fixing the project than getting more data; thus, the team never came across this ethical problem.

The limitations of this project included the time and resources. The computational power and speed of the entire pipeline, even using the faster runtime type on Google Colab, would take one hour to an

hour and a half to process through a roughly 250-frame scenario. The faster runtime type also meant that a compute unit had to be purchased just to allow for faster testing, allowing for more refining. The same limitation is also applied to the Gemini advice, as a budget had to be paid before it could be prompted for answers. Another limitation was the Qwen model used, as there are more capable Qwen models, but these other models require payments, like Google Gemini.

5. Conclusion

5.1. Future Work Recommendations

Many future works and recommendations could be made in this project. One future work may be to use other datasets such as the HRI Advice Dataset (HAD) [36] or the HRI Driver Behaviour Dataset (HDBD) [37], both of which are from the Honda Research Institute. Another future work would be the upgrading of the models used.

The improved Gemini model, Gemini 3.5 Flash, would have had improved response times as it excels in rapid loops involving complex coding [38]. The improved Qwen model, Qwen3-VL, would have improved spatial understanding, stronger multimodal reasoning, and improved visual perception and recognition [39].

Another possible future work would be to incorporate more input data using hardware, such as infrared cameras or Lidar, which would allow for improved SA understanding. One recommendation would be to have more testing undertaken on the prompt engineering and better prompts in general, mostly for the Gemini prompts.

Trying to get exact outputs was a hard challenge, and sometimes the advice given is still not great. Another recommendation is to look at upgrading some of the models used throughout the project, such as the Qwen or Gemini model, or completely switching a model out for another. This topic is very important as new laws in the USA require the use of monitoring cameras to detect if drivers are unfit to drive [40].

5.2. Key Findings

This report has demonstrated the successful development of a gaze-based SAE system capable of achieving Endsley's SA level 1(Perception) and partially achieving Level 2(comprehension) through the incorporation of gaze data and scene images. A major finding was the effective incorporation of VLMs into a real-time SAE system. Both Qwen2.5-VL and Gemini2.5-Flash were successfully integrated, with Qwen2.5-VL performing semantic analysis, constructing object relationships, and creating persistent knowledge graphs, whilst Gemini2.5-Flash processes and displays driving advice. These VLM integrations address a noteworthy research gap identified within the literature, where later approaches were usually limited to lower-performing models.

Another key finding is that VLMs were able to be easily integrated into an SAE system and work very efficiently and properly. This was a problem within other projects, as other projects could only use basic or low-benchmark scoring models or LLMs. With today's available models, better integration and use of VLMs was possible and completed. Another key finding is that there were other ways of calculating a driver's fixation and how long they were fixating for. In research, it was discovered that other projects used methods such as SGE and SAGAT to detect a fixated point, but not for how long. The fixation algorithm used in this project utilises the fixation point, the dispersion threshold before it is not considered a fixated point, and the minimum duration to be considered fixated.

The results further solidify that the combination of YOLO object detection, SAM2 segmentation, Qwen2.5-VL status, and relationship modelling and Gemini2.5-Flash driving advice provides useful and consistent outputs across different driving scenarios. These scenarios include urban, residential, sunny, overcast, snowy, daytime, and nighttime. The system successfully managed temporal consistency in the knowledge graph, whilst the application, the driving coach, generated actionable and relevant advice, validating the practicability of dynamic, gaze-informed knowledge graphs for SA monitoring. These findings will contribute to the improvement of IVMSs and emphasise the potential for such technology to exist in assisting safer transitions in driving environments.

6. References

- [1] M. R. Endsley, "Toward a Theory of Situation Awareness in Dynamic Systems," *Human Factors The Journal of the Human Factors and Ergonomics Society*, vol. 37, no. 1, 1995.
- [2] W. Ding, Y. Murzello, S. Cao and S. Samuel, "Exploring the Relationship Between Drivers' Stationary Gaze Entropy and Situation Awareness in a Level-3 Automation Driving Simulation," in *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 2024.
- [3] Y. Jiang, Q. Xu, K. Zhen, S. Wang and Y. Chen, "QUANTITATIVE EVALUATION OF DRIVER'S SITUATION AWARENESS IN VIRTUAL DRIVING THROUGH EYE TRACKING ANALYSIS," arxiv, 2024.
- [4] M. Hofbauer, L. Puttner, E. Steinbach, C. B. Kuhn and G. Petrovic, "Measuring Driver Situation Awareness Using Region-of-Interest Prediction and Eye Tracking," in *IEEE International Symposium on Multimedia (ISM)*, Naples, 2020.
- [5] A. Biswas, P. Gupta, S. Khurana, D. Held and H. Admoni, "Modeling Drivers' Situational Awareness from Eye Gaze for Driving Assistance," in *8th Conference on Robot Learning*, Munich, 2024.
- [6] C. Y. Arias-Portela, J. Mora and M. Caro, "Situational Awareness Assessment of Drivers Boosted by Eye-Tracking Metrics: A Literature Review," *Applied Sciences*, 2024.
- [7] F. Aderinto, J. B. Smith, M.-R. Giolando, P. Baskaran and J. A. Adams, "Improving Human-Robot Team Transparency with Eye-tracking based Situation Awareness Assessment," in *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, Boulder, 2024.
- [8] H. Zhou, Y. Lu and S. Fu, "The Assessment of Aviation Situation Awareness Based on Objective Behavior Methods," in *Engineering Psychology and Cognitive Ergonomics*, 2022.
- [9] M. Barz and D. Sonntag, "Automatic Visual Attention Detection for Mobile Eye Tracking Using Pre-Trained Computer Vision Models and Human Gaze," *Sensors*, 2021.
- [10] D.-Y. Cho and M.-K. Kang, "Human gaze-aware attentive object detection for ambient intelligence," *Engineering Applications of Artificial Intelligence*, vol. 106, 2021.
- [11] Q. Liu, R. Yu, Y. Cai, Q. Yuan, H. Wei and C. Lv, "Collision risk prediction and takeover requirements assessment based on radar-video integrated sensors data: A system framework based on LLM," *Accident Analysis and Prevention*, vol. 218, 2025.
- [12] J. Chen, K. P. Seng, J. Smith and L.-M. Ang, "Situation Awareness in AI-Based Technologies and Multimodal Systems: Architectures, Challenges and Applications," *IEEE Access*, vol. 12, pp. 88779-88818, 2024.
- [13] G. J. W. Xu, S. Pan, P. Z. H. Sun, K. Guo, S. H. Park, F. Yan, M. Gao, X. Wanyan, H. Cheng and E. Q. Wu, "Human-Factors-in-Aviation-Loop: Multimodal Deep Learning for Pilot

- Situation Awareness Analysis Using Gaze Position and Flight Control Data,” *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, vol. 26, no. 6, pp. 8065-8077, 2025.
- [14] H. Chen, X. Zhao, Z. Li, Q. Fu, Q. Wang and L. Zhao, “Construction and Analysis of Driver Takeover Behavior Modes Based on Situation Awareness Theory,” *IEEE TRANSACTIONS ON INTELLIGENT VEHICLES*, vol. 9, no. 2, pp. 4040-4054, 2024.
- [15] J. Lachance-Tremblay, Z. Tkiouat, P.-M. L’eger, A.-F. Cameron, R. Titah, C. K. Coursaris and S. S’en’ecal, “A gaze-based driver distraction countermeasure: Comparing effects of multimodal alerts on driver's behavior and visual attention,” *International Journal of Human-Computer Studies*, vol. 193, 2025.
- [16] C. Zhang, L. Hong, D. Wang, X. Liu, J. Yang and Y. Lin, “Research on Driving Scenario Knowledge Graphs,” *Applied Sciences*, 2024.
- [17] L. Halilaj, I. Dindorkar, J. L’uttin and S. Rothermel, “A Knowledge Graph-based Approach for,” in *European Semantic Web Conference*, 2021.
- [18] “LBW Dataset,” 2022. [Online]. Available: <https://drive.google.com/drive/folders/1nAHBb7Lg98x1aYdsaeM55XqFypIM8m7f>. [Accessed 09 18 2025].
- [19] “MEDIRL-EyeCar,” 2020. [Online]. Available: <https://github.com/soniabae/MEDIRL-EyeCar/tree/master/EyeCar>. [Accessed 15 08 2025].
- [20] “DADA2000,” 2024. [Online]. Available: https://drive.google.com/drive/folders/111_xOMWfs2eSoh0771ZJOcS2tcKwhh-C. [Accessed 15 08 2025].
- [21] “dreyeye,” 2019. [Online]. Available: <https://github.com/ndrplz/dreyeye?tab=readme-ov-file>. [Accessed 15 08 2025].
- [22] I. Kasahara, S. Stent and H. S. Park, “Look Both Ways: Self-Supervising Driver Gaze Estimation and Road Scene Saliency,” in *Computer Vision ECCV*, Tel Aviv, 2022.
- [23] Meta AI, “Segment Anything,” Meta AI, [Online]. Available: <https://segment-anything.com/>. [Accessed 18 10 2025].
- [24] Ultralytics, “Ultralytics YOLO Docs,” Ultralytics, [Online]. Available: <https://docs.ultralytics.com/>. [Accessed 18 10 2025].
- [25] “How does YOLO work?,” Tencent Cloud, 31 03 2025. [Online]. Available: <https://www.tencentcloud.com/techpedia/103581>. [Accessed 18 10 2025].
- [26] M. Kolek, “Fine-Tuning YOLO for Road Sign and Object Detection on Polish roads,” Medium, 07 08 2024. [Online]. Available: <https://medium.com/@mikolaj.kolek/fine-tuning-yolo-for-road-sign-and-object-detection-on-polish-roads-71a366e9a876>. [Accessed 14 09 2025].
- [27] Alibaba Cloud, “Qwen LLMs,” Alibaba Cloud, 11 09 2025. [Online]. Available: <https://www.alibabacloud.com/help/en/model-studio/what-is-qwen->

llm#:~:text=Qwen%20is%20a%20series%20of,range%20of%20fields%20and%20tasks..
[Accessed 18 10 2025].

- [28] Qwen Team, Alibaba Group, “Qwen2.5-VL Technical Report,” 19 02 2025. [Online]. Available: <https://arxiv.org/pdf/2502.13923>. [Accessed 18 10 2025].
- [29] R. D. Caballar and C. Stryker, “What is Google Gemini?,” IBM, [Online]. Available: <https://www.ibm.com/think/topics/google-gemini>. [Accessed 19 May 2026].
- [30] C. Moreira, J. Cockburn and M. S. Castelhano, “A Framework for Leveraging LLMs for Scene Analysis and Cognitive Processing,” *Proc. ACM Comput. Graph. Interact. Tech*, vol. 8, no. 2, 2025.
- [31] D. D. Salvucci and J. H. Goldberg, “Identifying Fixations and Saccades in Eye-Tracking Protocols,” in *ETRA '00: Proceedings of the 2000 symposium on Eye tracking research & applications*, Palm Beach Gardens, 2000.
- [32] M. Costa, A. Simone, V. Vignali, C. Lantieri and N. Palena, “Fixation distance and fixation duration to vertical road signs,” *Applied Ergonomics*, vol. 69, pp. 48-57, 2018.
- [33] T. Foulsham, E. Walker and A. Kingstone, “The where, what and when of gaze allocation in the lab and the natural environment,” *Vision Research*, vol. 51, no. 17, pp. 1920-1931, 2011.
- [34] J. M. Henderson, “Human gaze control during real-world scene perception,” *Trends in Cognitive Sciences*, vol. 7, no. 11, pp. 498-504, 2003.
- [35] Engineers Australia, “Code of Ethics and Guidelines on Professional Conduct,” 08 2022. [Online]. Available: <https://www.engineersaustralia.org.au/sites/default/files/2022-08/code-ethics-guidelines-professional-conduct-2022.pdf>. [Accessed 19 10 2025].
- [36] Honda Research Institute, “HAD HRI Advice Dataset,” Honda Research Institute, 2019. [Online]. Available: <https://usa.honda-ri.com/had>. [Accessed 23 May 2026].
- [37] Honda Research Institute, “HDBD HRI Driver Behavior Dataset,” Honda Research Institute, 2022. [Online]. Available: <https://usa.honda-ri.com/hdbd>. [Accessed 23 May 2026].
- [38] Google, “Gemini 3.5 Flash,” Google, [Online]. Available: <https://ai.google.dev/gemini-api/docs/models/gemini-3.5-flash>. [Accessed 22 May 2026].
- [39] QwenTeam, “Qwen3-VL: Sharper Vision, Deeper Thought, Broader Action,” Qwen, 23 September 2025. [Online]. Available: <https://qwen.ai/blog?id=qwen3-vl>. [Accessed 22 May 2026].
- [40] A. Alaniz, “Your Car May Soon Be Monitoring Everything You Do Behind The Wheel,” Motor1, 28 April 2026. [Online]. Available: <https://www.motor1.com/news/794316/new-cars-mandate-impaired-driver-detection-delay/>. [Accessed 23 May 2026].
- [41] Meta AI, “Segment Anything Model 2,” Meta AI, [Online]. Available: <https://ai.meta.com/sam2/>. [Accessed 18 10 2025].

- [42] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar and R. Girshick, “Segment Anything,” Meta AI Research, 2023.
- [43] A. W, “YOLO Object Detection Algorithms 101: Part 1,” basicAI, [Online]. Available: <https://www.basic.ai/blog-post/yolo-object-detection-algorithms-101-part-1#:~:text=in%20Object%20Detection-,What%20is%20YOLO?,passes%20or%20two%2Dstep%20processes..> [Accessed 18 10 2025].
- [44] Google, “Gemini 2.5 Flash-Lite,” Google, [Online]. Available: <https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash-lite>. [Accessed 20 May 2026].
- [45] Google, “Gemini thinking,” Google, [Online]. Available: <https://ai.google.dev/gemini-api/docs/thinking>. [Accessed 20 May 2026].
- [46] Google, “Google Colaboratory,” Google, [Online]. Available: <https://colab.google/>. [Accessed 18 10 2025].
- [47] Meta AI, “Detectron2,” Meta AI, [Online]. Available: <https://ai.meta.com/tools/detectron2/>. [Accessed 19 10 2025].
- [48] Meta AI, “Llama 3,” Meta AI, [Online]. Available: <https://www.llama.com/models/llama-3/>. [Accessed 19 10 2025].
- [49] J. Kim, T. Misu, Y.-T. Chen, A. Tawari and J. Canny, “Grounding Human-To-Vehicle Advice for Self-Driving Vehicles,” in *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

7. Appendices

7.1. Appendix A: Research Project Timeline

#	Focus	Deliverable	Dependency	Milestone	Final Notes
1	Proposal	Literature Review & research Proposal		2025, Semester 2, Week 7	Was submitted on time.
2	Initial testing & Evaluation	Conceptual Evaluations based on commercially available models	1	2025, Semester 2, Week 8	Testing methods and capabilities of Object Detection and Segmentation as well as Feature Extraction
3	Conceptual Design draft	Conceptual design draft	2	2025, Semester 2, Week 9	Testing methods and capabilities of Object Detection and Segmentation as well as Feature Extraction and LLM's & MLLM's
4	Finalise Conceptual Design	Conceptual design	3	2025, Semester 2, Week 11	Finalise a version utilising feature extraction, object detection, segmentation, MLLM's for object relations and knowledge graph generation. An MLLM for SA scoring was completed in 2025, Semester 2, Week 12
5	Interim Report & Presentation	Progress Report & Presentation	3&4	2025, Semester 2, Week 13	Was submitted on time
6	Review	Review meeting Possible change of testing and validation (may need UHREC approval)	5	2026, Semester 1, Week 2	A quick review of progress made so far. Problems were identified and were fixed during the second half of the project. UHREC approval was not need as no simulated testing was being done.
7	Feature Extraction	Reprocessed Gaze Dataset and Feature sets	6	2026, Semester 1, Week 3	Was a part of the conceptual stage, worked on around 2025, Semester 2, Week 8

8	Build/Construct	Finalise model for SA scoring model	7	2026, Semester 1, Week 4	Was a part of the conceptual stage, worked on around 2025, Semester 2, Week 12
9	LLM & MLLM	LLM or MLLM utilised SAE module	8	2026, Semester 1, Week 6	Was a part of the conceptual stage, worked on around 2025, Semester 2, Week 11
10	Object Detection and Segmentation	Integrate object detection and segmentation	9	2026, Semester 1, Week 7	Was a part of the conceptual stage, worked on around 2025, Semester 2, Week 8
11	Test & Validate / Verify	Simulation scenarios testing and evaluation	9 & 10	2026, Semester 1, Week 8	Was a part of the final stage, worked on around 2026, Semester 1, Week 10 & 11
12	Draft Report	Draft of final report		2026, Semester 1, Week 11	
13	Project Delivery & Oral Defence	Final Project Report, Project defence presentation, Final SAE system.	11	2026, Semester 1, Week 13	

7.1. Appendix B: SAM2 Justification Table

Reason	Discussion	Justification
Versatility	SAM has been trained on 11 million images, which also includes 1.1 billion masks, whilst SAM2 has been trained on the same data, as well as roughly 51 thousand videos, including 600 thousand masks, along with the video data [23] [41]	The SAM2 model works upon SAM, allowing for videos and images, which will be useful. Input images will be able to be processed to a greater extent, whilst also being able to process video data, if required.
Generalisation	Being trained on such a big dataset allows SAM to define objects. This massive amount of training grants zero-shot generalisation to unknown images and objects without additional training. [23]	This will be extremely useful when analysing driving images, as most of these images will include unknowns.
Efficiency & Scalability	SAM2 offers faster inference times and superior handling of dynamic scenes, such as videos or multiple images, whilst being able to use the same prompt inputs as SAM. [41]	This will be useful as this system will be required to process many masks across the images of the dataset, allowing for an efficient process. It may also be required to process masks across videos, which SAM2 processes better than SAM.
Integration	SAM prompt-based approach enables for integration with fixation data (x, y), allowing segmentation of specific regions without the retraining the model. [42] This feature also carries over to SAM2 and may even be possible with videos and fixation data. SAM2 is even able to track objects that temporarily exit the frame, as it has context from previous frames. [41]	This will be useful as SAM2 will be able to segmentate a single fixation point, allowing for an object to be identified as the object the driver may be looking at.

7.2. Appendix C: YOLO Justification Table

Reason	Discussion	Justification
Real-time Object Detection capability	YOLO uses a single-stage object detection method, meaning it only looks at the image once, whilst other models will take multiple looks at the same image. It can also process up to an impressive 45 frames per second [43].	This would be beneficial in a driving scenario as this model would be able to process images in a fast and efficient manner, supporting frame-by-frame object detections.
High Accuracy & Robustness	The pretrained YOLO model being used was trained using driving and traffic datasets, as well as having driving and traffic-specific classes [26].	This would significantly help with driving specific object detections, as it will not have objects being misclassified, such as a traffic sign being mistaken for an animal (YOLOv11). This pretrained YOLO model would not have classes such as animals as part of its model.
Integration	YOLO models provide many different prompts, such as bounding boxes along with class labels.	This would significantly help with how the prompts for the VLM, along with the segmented gaze point, would also have detected other traffic-significant objects within the image, allowing for the VLM to effectively be used.

7.3. Appendix D: QWEN Justification Table

Reason	Discussion	Justification
Visual Perception & Resolution Handling	This model stands out in visual perception as it is allowed to analyse images of differing resolutions. It also supports multi-image reasoning for comparing and evaluating images [28].	This will be incredibly useful as unseen objects, such as faraway cars, may be picked up by the VLM. It will also be beneficial when it comes to creating accurate perceptions or relationships between certain objects detected.
Efficiency and Scalability	Qwen2.5-VL offers multiple different versions, with differing use cases (3B, 7B & 72B). The model being used in this case is the 7B version, as it is considered the high-end personal used version. But if an enterprise-size version is needed, they do provide the 72B version as well [28].	The 7B version is used as it provides a balance between capability and efficiency. If longer datasets or videos were to be used, the 72B version would have been considered.
Versatility	Qwen2.5-VL supports multiple input prompts such as images, URLs, and text. It is capable of handling different tasks, such as multi-modal reasoning and creating relationships between multiple objects within an image [28].	The model design, in combining gaze-mapped data, segmentation, object detection, and text-based instruction prompts, helps to produce the required outputs needed. This will significantly help in developing attention-driven knowledge graphs.

7.4. Appendix E: Gemini Justification Table

Reason	Discussion	Justification
Speed and cost efficiency	This model stands out as the most cost efficient multimodal model, offering the fastest performance for high-frequency, lightweight tasks. It is also best for high-volume classification, simple data extraction, and extremely low-latency applications where budget and speed are the primary constraints [44].	This will be extremely useful as in a real-case scenario, advice and SA scores would need to be created very quickly and money would need to be used sparingly.
Versatility and generalisation	With it being a multimodal model, taking images and texts in different arrangements, it is also capable of understanding and reasonings, allowing for better responses. These inputs are usually unknown or unseen images, which this model has great at dealing with [45].	This will be crucial as the advice given will need to be very thorough and to the point, whilst communicating the correct advice and not irrelevant advice. With the model being multimodal and having great generalisation, the required inputs, such as knowledge graphs and gaze data are easily interpreted by Gemini.

7.5. Appendix F: Concept 1 (SAM / YOLO)

The 1st conceptual approach involved evaluating the combined use of SAM and YOLO for segmentation and object detection. SAM was applied to the full image using the 'SAMAAutomaticMaskGenerator' function without any augmentation, which can be seen in Figure 7F.1. The masked image was then passed into the YOLO model for object detection. This conceptual method proved invalid, as it did not involve the driver's gaze, a crucial part of the project. The YOLO model also produced detections with poor confidence, which can be seen in Figure 7F.2.



Figure 7F.7.1: SAM with no Augmentations



Figure 7F.7.2: SAM with no Augmentations and YOLO detections

Consequently, the dataset was filtered to a more appealing subject dataset for initial testing. The Non-Maximum Suppression was also not implemented, allowing for overlaid detections. (Code for the SAM segmentation can be seen here https://colab.research.google.com/drive/1mO-X_w1W-bXQbnmjyiDdVvP3lOPMnijC?usp=sharing).

7.6. Appendix G: Concept 2 (SAM + YOLO)

Following a review of the previous approach, a new method was developed to compare two processes. One standalone YOLO version and a combined SAM + YOLO version. The approach executes both methods individually: the YOLO-only version applies the pre-trained YOLOv8 model to generate a baseline set of detections. The SAM + YOLO version uses SAM to create masks, crops the original image to each mask's bounding box, and runs YOLO on the cropped sections. Detections common to both versions are considered fusions. The results are compared visually by overlapping bounding boxes on the image seen in Figure 7G.1, with blue indicating 'YOLO-only', red indicating 'SAM + YOLO,' and green indicating fusion.



Figure 7G.7.3: Conceptual Stage 2 Results

a). Snowy Weather

b). Overcast Weather

c). Sunny Weather

As shown in Figure 7G.1.a, both methods detected largely the same objects, with the 'SAM + YOLO' version identifying an additional object missed by the 'YOLO-only' version. In contrast, Figures 7G.1.b and 7G.1.c show low fusion detections, with 'SAM + YOLO' tending to over-detect objects. Unlike the first concept, this approach used a pre-trained YOLOv8 model and an SAM with modified augmentations to avoid unwanted segmentation. Despite the fact that the approach still did not involve the driver's gaze data, it illustrates enhanced detection of driving-related objects. The results of this concept indicate that combining the models enhances performance.

(Code for this conceptual stage can be seen here

https://colab.research.google.com/drive/1xVKivxGaYywnVSIgh3_JRPahEl6NOsG9?usp=sharing)

7.7. Appendix H: Concept 3 (Qwen + SAM + YOLO)

The third conceptual stage draws inspiration from multiple papers, particularly ‘A Framework for Leveraging LLMs for Scene Analysis and Cognitive Processing’. [30] This approach follows a 6-step process, concluding with a work-in-progress evaluation stage. This pipeline process can be seen in Figure 7H.4 below.

- Step 1). Set up and Data Loading
- Step 2). Gaze Data Processing and Fixation Detection
- Step 3). Mapping fixation to images
- Step 4). Generate gaze prompts for segmentation
- Step 5). Model Loading (pre-train YOLOv8, SAM, Qwen2.5-VL and DSPy modules)
- Step 6). Main Processing Loop
- Step 7). Evaluation method

Figure 7H.7.4: Conceptual Stage 3 Method Process

The most important, step 6, executes the main processing loop, iterating through each image via a for loop. The operations performed within the loop can be visualised in Figure 7H.5.

- Loop Step 1). Loads image
- Loop Step 2). Run pretrained YOLOv8 for object detection, getting bounding boxes, classes and confidence scores.
- Loop Step 3). Apply SAM2 to create segmentation masks using the YOLOv8 bounding boxes and gaze prompts (See Appendix K for output example).
- Loop Step 4). Uses ‘TemporalObjectTracker’ to connect current detections with previous scenes and preserve consistent object IDs.
- Loop Step 5). Utilise VLM via ‘TemporalObjectLabeler’ to clarify object labels using the image, YOLO detections, and tracking history.
- Loop Step 6). Update the knowledge graph using the ‘TemporalGraphUpdater’, including new objects and their spatial relationships.
- Loop Step 7). Save detections, segmentation, labelling and graph data for current image.
- Loop Step 8). Generate an output visualisation displaying the scene with bounding boxes, object labels and confidence scores, scene description and the updating knowledge graph. An example of this output can be seen in Figure 7H.6.

Figure 7H.7.5: Conceptual Stage 3 Main Loop Process Steps

This evaluation step evaluates the quality of the knowledge graphs generated by the previous pipeline step. Following the supervisor’s advice, a few-shot learning method was used. A dedicated evaluation module was developed that utilises another instance of the Qwen VLM through manual prompting. The prompt includes few-shot examples, the current image, scene description and generated knowledge graph. The VLM then evaluates the knowledge graphs accuracy and outputs a score along with reasoning.

Code for this current method can be here: (https://colab.research.google.com/drive/1CkZQ-aQgLm6DnG89y_sIPOrE-lqKlmpQ?usp=sharing).

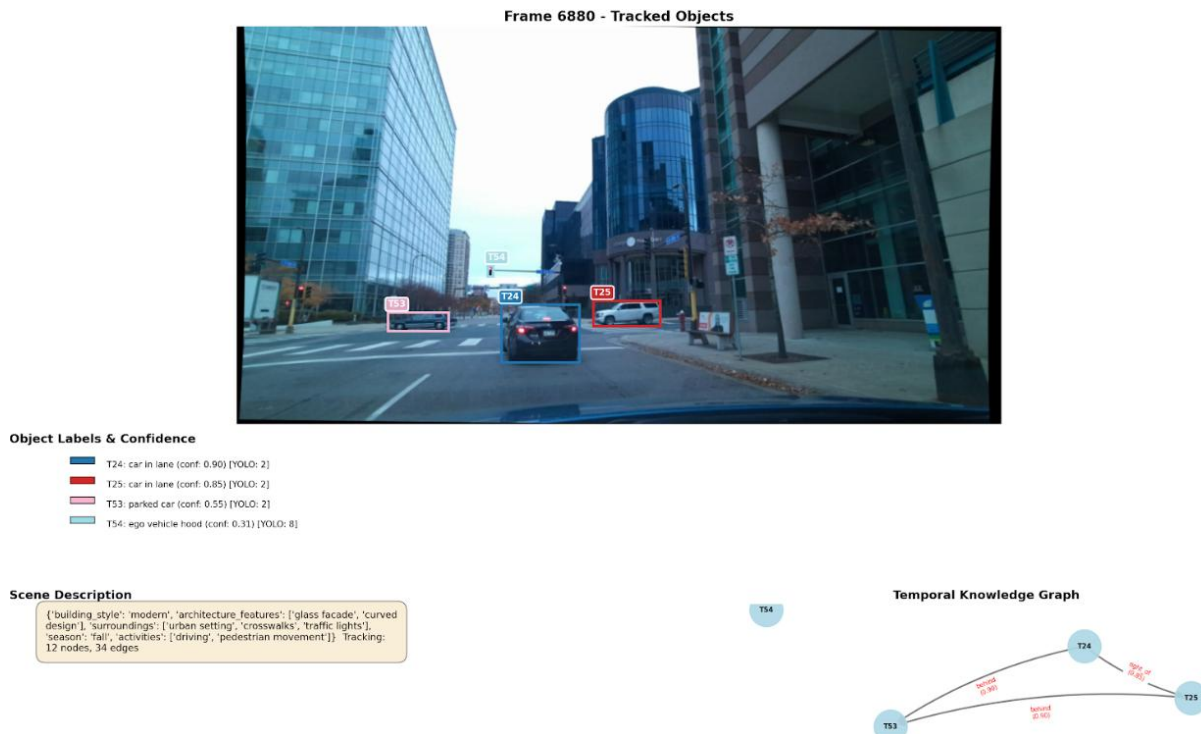


Figure 7H.7.6: Frame Analysis Example

One section inspired by the paper ‘A Framework for Leveraging LLMs for Scene Analysis and Cognitive Processing’ [30] could not yet be implemented due to integration challenges, this section involved testing alternative gaze prompt configurations for SAM2 (pointwise, circle, triangle and cross) to capture more information around the fixation point. After evaluation on multiple images, the default point-wise prompt was kept as the most suitable. Although alternative shapes produced higher internal SAM scores, they frequently segmented overly large or irrelevant regions, such as a car and entire road or buildings

7.8. Appendix I: SAM & SAM2 Process



7.9. Appendix J: Final Methodology Code

https://colab.research.google.com/drive/1aYR4o_UBao4Gkn5yJgwJqimnhc92wil0?usp=sharing

7.10. Appendix K: Evaluation Spreadsheet

<https://docs.google.com/spreadsheets/d/1DkqEnb7IZVgqr6Buvj8DK95furTb8IUg/edit?usp=sharing&ouid=115751822192706699163&rtpof=true&sd=true>

7.11. Appendix L: 7 Scenarios Files

<https://drive.google.com/drive/folders/13RfnyVADN7A1xiIsKmchiIo3PeI-nhAq?usp=sharing>

7.12. Appendix M: Subject 17-3, Frame 101

